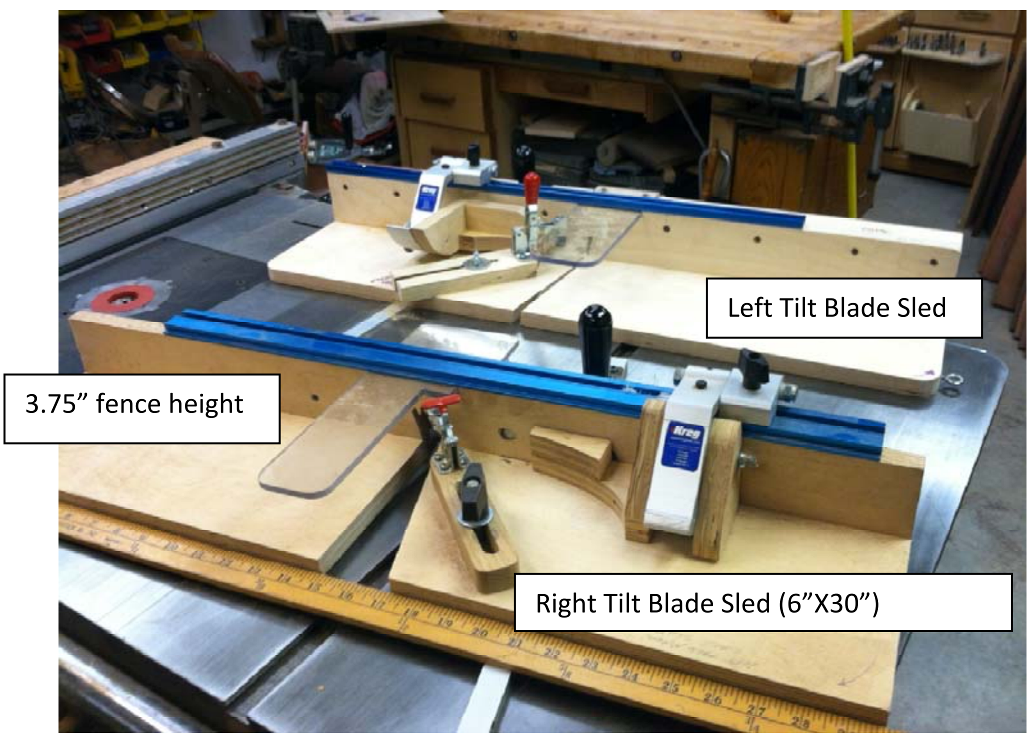
Table Saw Miter Sled Construction
by
Jay S. Helland

Safety and Hold Harmless Statement: Woodworking can be very dangerous! The person choosing to construct this sled takes full responsibility for its proper use and any injuries that may occur in its construction or use. Be sure you read and understand the manufacturer's instructions that come with your tools before using them.

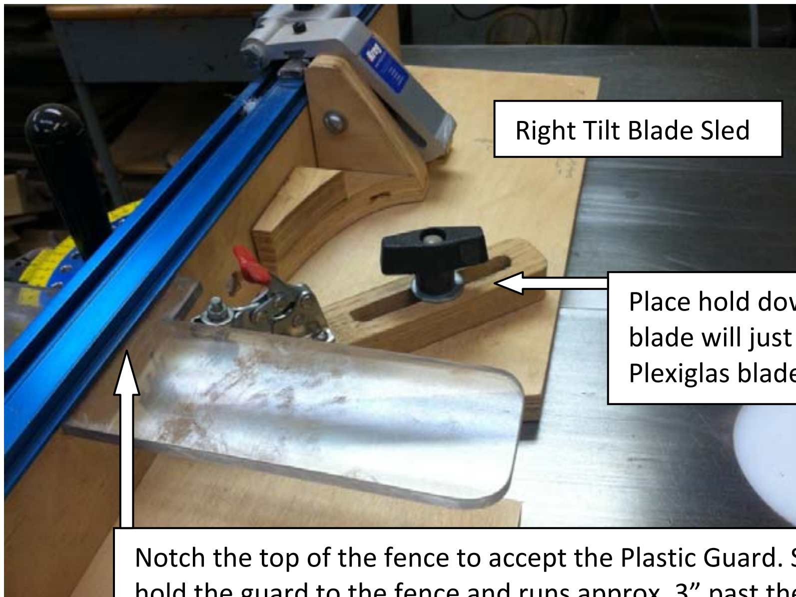
The following photos are provided so that those wishing to fabricate this type of sled have a visual image of how the sled was constructed. It is assumed that the craftsman has the skills and experiecne to deduce from the photos what construction methods were used. It may necessary to make slight modifications to the design to fit your table saw. Please pay close attention to the captions associated with the photo so that you are looking at the correct blade tilt. If you decide <u>not</u> use a flip stop mounted to the top of the fence and instead use a clamp and wooden stop, you obviously would not need to purchase and install the top flip stop and rail system.

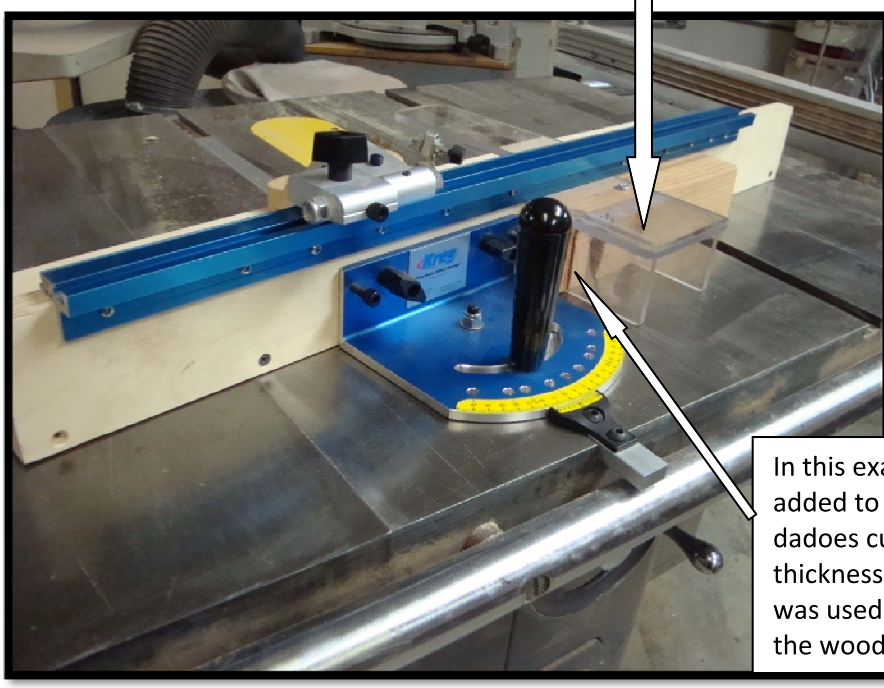
The after-market miter gauge used must have the ability to divide an angle into tenths of a degree. Currently, Kreg and Incra are the two afer-market miter sled manufacturers that produce a veriner scale on their miter guages. Others may exist but the author is not aware of any at this time.

Tools and Materials Needed: table saw, drill press, screwdrivers, wrenches, machine screws that fit the holes in the head of the miter guage, transfer punches (see photo), ¾" baltic birch plywood, wood screws, fender washer and toggle holddown clamp.



The two sleds shown have been a work-in-progress and are not identical to each other. Both sleds function in the same but through the continuous-improvement process they are always being modified for increased functionality. Likewise, use skills you have to make improvements to this design.

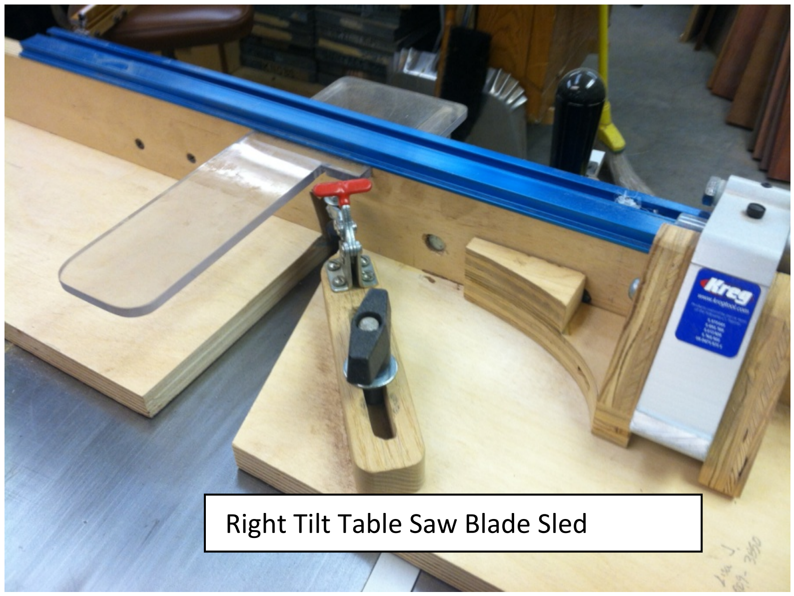




The view above shows how the Kreg Miter Gauge is attached to the Baltic Birch plywood used to carry the segments. This sled is for a right titling blade table saws (photo: Delta 1952 Unisaw). A left titling blade will use the right miter slot in the table saw top.

- Used butt joints reinforced with wood screws.
- Use a transfer punch to locate the center of the hole in the head of the miter gauge for the plywood fence.
- Drill 1/16” pilot holes using a drill press in the plywood fence.

- Turn the fence over counter bore the holes large enough to fit a socket on to the bolt head or use a carriage bole. Either way the bolt heads need to be flush or just below the surface of the vertical fence.
- Attach the fence to the miter gauge. Place approximately 1/3 or 10 to 15” of the fence on the side of the flip stop.
- Clamp the plywood that carries the wood to be cut into segments to the vertical fence using a butt joint and wood screws.

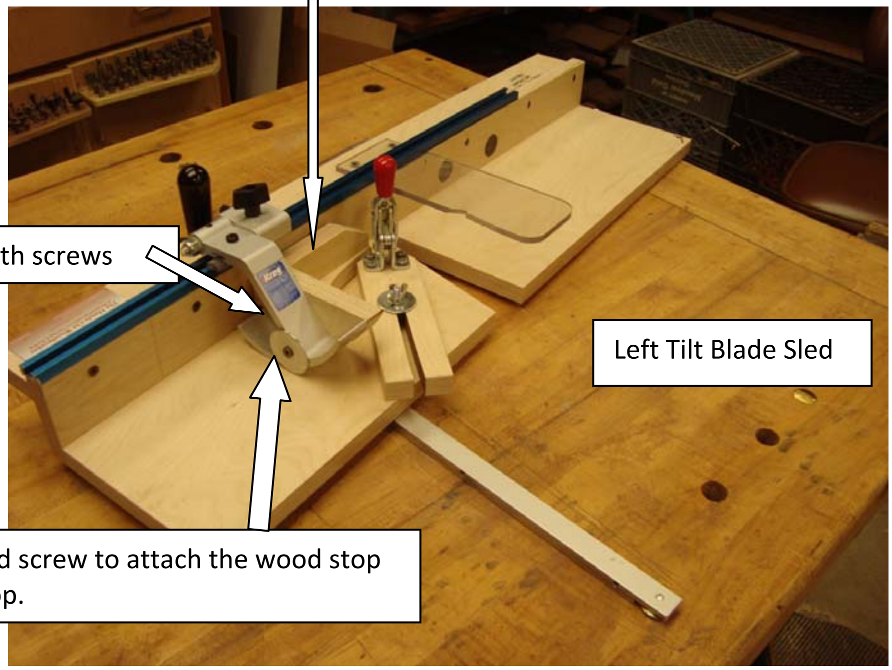


Right Tilt Table Saw Blade Sled

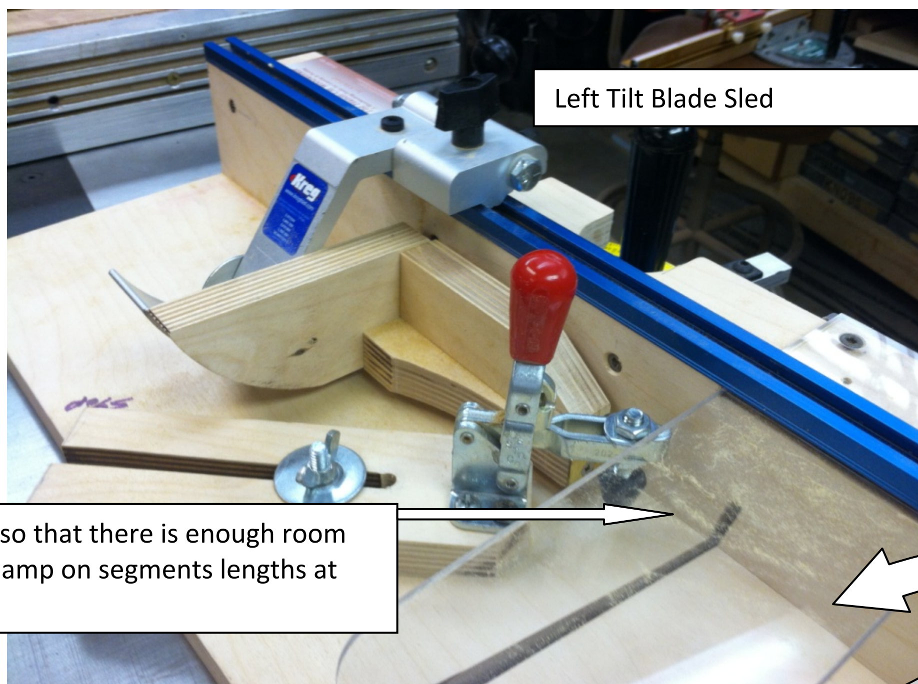
The rear blade guard can be fabricated from Plexiglas or wood. It is recommended that a magnet stop be clamped to the table saw to prevent the blade from going past the rear guard.

The extended wood stop is needed to make room for the hold-down clamp.

Attach extended wood stop with screws



Use a fender washer and screw to attach the wood stop to the aluminum flip stop.



Offset the Blade Guard so that there is enough room to use the hold-down clamp on segments lengths at least 1” wide.

The front blade guard should be made of Plexiglas. When cutting wood segments to width, you should **never reach under the guard with your fingers.**

Transfer Punch Set